## PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

## ARTICLE DETAILS

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| **TITLE (PROVISIONAL)** | Examining LGBTI+ Inclusive Sexual Health Education from the perspective of both youth and facilitators: A Systematic Review |
| **AUTHORS** | O'Farrell, Muire; Corcoran, Phil; Davoren, Martin P. |

## VERSION 1 – REVIEW

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| **REVIEWER** | Carrie Llewellyn<br>Brighton & Sussex Medical School, Primary Care & Public Health |
| **REVIEW RETURNED** | 27-May-2020 |

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| **GENERAL COMMENTS** | The topic is an important and interesting one. Thank you for allowing me to review this paper. The writeup reads well, however there are a number of methodological concerns I have outlined below.<br><br>Considerations:<br>In the title plse indicate who the education training is aimed at.<br><br>Introduction: 3rd paragraph - plse elucidate or rephrase 'lack of language or unwillingness to use said language' as it is not clear what is meant by this.<br><br>It is not clear how 'young people' is defined for the purposes of this paper as age appropriate educational materials will vary according to age, setting etc. Ditto in the search terms.<br><br>Of more major consideration is the overall aim of the review 'to synthesise the data available in relation to inclusive evidence based sexual health education and training for LGBT+ young people'. This is very broad/vague and is in part responsible for the lack of definition in the search terms. It is not clear whether this review is seeking to ask young people what they think about sex education they have received or whether it is seeking to find evidence based training resources and materials in order to deliver sex education to young people. This confusion has lead to a mixture of studies being presented in the review. Some of the studies included contain the opinions of participants about relationships (e.g. Donovana & Hester) and some contain studies testing feasibility and acceptability of different sexual health promotion resources (e.g. Mustanski et al, Buston & Hart). This is a result of the vagueness over the aim (which does not quite represent the title as is) and the eligibility criteria.<br><br>A clearer rationale for the study and objectives defined using PICO may help define the specific questions of interest.<br><br>The inclusion of Medline may have increased the comprehensiveness of the search. |

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| | There is no indication of how many authors were involved in the stages of the review i.e. identification, screening, eligibility and quality appraisal phases. It appears that only one author was involved in all stages (the researcher approached the date, the researcher individually coded...'). Normal practice would be to have more than one independent coder involved in varying stages to lend credibility to the validity of the methods/findings.<br><br>The findings are a little long and could be edited - again more specific aims, research questions would help guide this. |

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| **REVIEWER** | Michael Brown<br>Queen's University Belfast |
| **REVIEW RETURNED** | 15-Jul-2020 |

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| **GENERAL COMMENTS** | This is a potentially important systematic review on a contemporary subject area in need of inquiry and analysis. There are some issues that the authors are invited to consider and address to improve their paper.<br><br>1. Page 4, line 32, please provide details and examples of the differences referred to.<br>2. Suggest including a statement that ethics review and approval not required for the literature review.<br>3. Page 6, PRISMA figure, please provide details of the reasons why the full text articles were excluded<br>4. Suggest including an additional column that presents the key findings from the included studies. Please also provide more specific details in the methods/data gathering column of the table of the data collection method used in each study. Please also add in details of the countries where the studies were undertaken in the first column.<br>5. Page 12, line 56 typographical error<br>6. Page 15, second paragraph, please include additional details of the analysis process and who was involved and how confirmation was achieved and biases minimised.<br>7. You provide a comprehensive overview of the included studies, however in at times it reads like a list rather than an integrated synthesis of the themes and findings - for example, page 19, from lines 38-55. Suggest further editing and synthesising of the themes would improve this part of the paper.<br>8. Please provide details of the facilitators on page 17, line 20 - who were they and what did they facilitate?<br>9. What were facilitators poorly equipped with? Knowledge in general or LGBT specifically - page 17, line 36<br>10. Please clarify the issues that caused uncertainty and anxiety - page 18, line 27<br>11. Please clarify the term, 'bottom surgery' on page 18, line 47<br>12. Please provide details regarding the use of or absence of theoretical models used in the programmes. Were they used and if so which ones and if not what were the programmes informed and built around regarding evidence of changing behaviours?<br>13. Page 19, line 19 - please provide details of the standardisation required within the programmes and what the studies recommended needs to be included<br>14. Page 20, line 51-53, please include reference behaviour changes models and theories or the absence.<br>15. Please review the discussion and include details of the implications for sexual health policy - what is missing and what needs to be developed and included based on the review? Similarly, what are the implications of the findings on practice and those |

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| | seeking to develop and deliver such programmes in the future?<br>Gaps in knowledge and confidence features in your review - what are the education and practice development needs that need to be addressed?<br>16. Please set out the gaps in research based on your analysis of the current evidence and the areas for future research and priorities. |

**VERSION 1 – AUTHOR RESPONSE**

Reviewer: 1

The topic is an important and interesting one. Thank you for allowing me to review this paper. The writeup reads well, however there are a number of methodological concerns I have outlined below.

Considerations:
In the title please indicate who the education training is aimed at.
• The Title has now been changed to 'Examining LGBTI+ Inclusive Sexual Health Education from the perspective of both youth and facilitators: A Systematic Review'. This is to represent both youth and youth serving professional perspectives, to create a well-rounded understanding of the issues experienced in both delivering and receiving inclusive sexual health education.

Introduction: 3rd paragraph - please elucidate or rephrase 'lack of language or unwillingness to use said language' as it is not clear what is meant by this.
• This has now been updated as reads as follows 'Research indicates that when LGBTI+ sexuality is included, it is regarded as largely unhelpful by LGBTI+ youth due to a lack of appropriate LGBTI+ terminology and sexual health specific language such as gender identity e.g. cisgender and transgender [6,7].'

It is not clear how 'young people' is defined for the purposes of this paper as age appropriate educational materials will vary according to age, setting etc. Ditto in the search terms.
• The term young people has now been defined within the paper. It reads as follows '- The term 'young people' for the purpose of this article refers to individuals aged between 16 years to 23 years. This age bracket represents the average ages for teenagers attending second and 3rd level education, also referred to as high school and college level education, who would be in receipt of sexual health education that covers sexual identity and sexual intercourse.'

Of more major consideration is the overall aim of the review 'to synthesise the data available in relation to inclusive evidence based sexual health education and training for LGBT+ young people'. This is very broad/vague and is in part responsible for the lack of definition in the search terms. It is not clear whether this review is seeking to ask young people what they think about sex education they have received or whether it is seeking to find evidence based training resources and materials in order to deliver sex education to young people. This confusion has led to a mixture of studies being presented in the review. Some of the studies included contain the opinions of participants about relationships (e.g. Donovana & Hester) and some contain studies testing feasibility and acceptability of different sexual health promotion resources (e.g. Mustanski et al, Buston & Hart). This is a result of the vagueness over the aim (which does not quite represent the title as is) and the eligibility criteria.
• Following discussion, the authors have reviewed the aim. This has been updated as reads as follows 'There is a need to understand best practice with regard to the delivery of LGBTI+ inclusive sexual health education [9]. Evidence is needed to develop strategies that improve access for youth and develop appropriate and comprehensive training for professionals. The aim of this review is to synthesise the data available in relation LGBTI+ sexual health education for young people aged 16-23

years from the prospective of LGBTI+ youth and the professionals who deliver sexual health education in order to understand how best this can be improved.'

A clearer rationale for the study and objectives defined using PICO may help define the specific questions of interest.
• Thank you for this suggestion. Following discussion, the authors utilised the PICO framework in framing our revised submission and redefining our aim.

The inclusion of Medline may have increased the comprehensiveness of the search.
• Many thanks for this insight. The inclusion of Medline may have increased the comprehensiveness of the search and is something the authors will take account of in the future. This has also been added as a limitation of the search strategy. It reads as follows 'The inclusion of Medline as a fourth search engine, may have increased the comprehensiveness of the search.'

There is no indication of how many authors were involved in the stages of the review i.e. identification, screening, eligibility and quality appraisal phases. It appears that only one author was involved in all stages (the researcher approached the date, the researcher individually coded...'). Normal practice would be to have more than one independent coder involved in varying stages to lend credibility to the validity of the methods/findings.
• In order to address issuing relating to the validity of the methods/findings this section was reviewed and action taken. This now reads as follows 'The results sections of each paper were analysed by all three authors to identify evidence of the impact of sexual health education in a variety of contexts. First, a preliminary synthesis was developed. The same relevant data were extracted from all papers and tabulated. A content analysis with an inductive approach was then conducted which allows themes to emerge from the data (See Table 2). Two co-authors (MOF and MPD) proposed and defined categories following analysis. Any discrepancies were discussed with a third author (PC).'

The findings are a little long and could be edited - again more specific aims, research questions would help guide this.
• Thank you for this insight. The results section has been edited and further synthesised to address this comment in line with the new aim. This has resulted in three more refined result categories-
1. Online Learning to Facilitate Accessible Equitable Sexual Health Education.
2. Inclusive Sexual Health Education- Programme Components
3. Defining Facilitator Attributes for Inclusive Learning

Reviewer: 2

This is a potentially important systematic review on a contemporary subject area in need of inquiry and analysis. There are some issues that the authors are invited to consider and address to improve their paper.

1. Page 4, line 32, please provide details and examples of the differences referred to.
• This now reads as follows 'As a result, the information provided to LGBTI+ youth varies greatly across educational settings based on the facilitators own values and comfort with the subject [6,8]. For example, where facilitators were embarrassed to make mistakes in relation to discussing LGBTI+ pronouns or explaining different genders such as non-binary, many chose not to talk about these topics at all [6,8].'

2. Suggest including a statement that ethics review and approval not required for the literature review.
• This has now been added to the search strategy and reads as follows 'Due to the nature of this type

of research, ethical approval and review was not required.'

3. Page 6, PRISMA figure, please provide details of the reasons why the full text articles were excluded
• The PRISMA figure has now been updated to provide details of the reasons full text articles were excluded

4. Suggest including an additional column that presents the key findings from the included studies. Please also provide more specific details in the methods/data gathering column of the table of the data collection method used in each study. Please also add in details of the countries where the studies were undertaken in the first column.
• An additional two columns have been added to Table 1. Summaries of Studies Included in the Review. One presents the key findings from the included studies, and one details the countries where the studies were undertaken. Additional data on the methods/data gathering has also been included in the relevant column in the table.

5. Page 12, line 56 typographical error
• This has been rectified

6. Page 15, second paragraph, please include additional details of the analysis process and who was involved and how confirmation was achieved and biases minimised.
• This section has now been updated as highlighted above

7. You provide a comprehensive overview of the included studies, however in at times it reads like a list rather than an integrated synthesis of the themes and findings - for example, page 19, from lines 38-55. Suggest further editing and synthesising of the themes would improve this part of the paper.
• Thank you for this insight. The results section was further synthesised and edited to reflect a more in-depth analysis of the main themes.

8. Please provide details of the facilitators on page 17, line 20 - who were they and what did they facilitate?
• The term facilitators has now been defined within the paper. It reads as follows 'The term 'facilitator' is used for the purpose of this article to refer to those delivering sexual health education. This encompasses teachers, educators, trainers and any form of professional who delivers sexual health education.'

9. What were facilitators poorly equipped with? Knowledge in general or LGBT specifically - page 17, line 36
• The results section of the paper has been further synthesised and edited altering the text this comment refers to.

10. Please clarify the issues that caused uncertainty and anxiety - page 18, line 27
• The results section of the paper has been further synthesised and edited altering the text this comment refers to.

11. Please clarify the term, 'bottom surgery' on page 18, line 47
• The term 'bottom surgery' has now been clarified and reads as follows 'youth needed more in-depth information around gender affirming interventions both medical and non-medical e.g. binding and 'bottom surgery' (this is a form of gender affirmation surgery for a transfeminine person).'

12. Please provide details regarding the use of or absence of theoretical models used in the programmes. Were they used and if so which ones and if not what were the programmes informed

and built around regarding evidence of changing behaviours?
• This review did not examine the theoretical models used in the programmes, as this study was focused on the experience and/or opinions of inclusive education from the perspective of professionals and LGBTI+ young people. Not all studies included were programme evaluations or included programme details. However, this is an area we consider important for future research and a limitation of this review. We have included it as such. It reads as follows 'This review did not account for the use of or absence of theoretical models within sexual health education programmes and their ability to change behaviours, something that should be considered in future research.'

13. Page 19, line 19 - please provide details of the standardisation required within the programmes and what the studies recommended needs to be included
• The section the reviewers comment addresses has since been removed and the results section rewritten. The subject addressed by this comment is now addressed in one of the main themes discussed in the results section. It is titled '3.3 Inclusive Sexual Health Education- Programme Components'

14. Page 20, line 51-53, please include reference behaviour changes models and theories or the absence.
• The section the reviewers comment addresses has since been removed as the results section has been rewritten.

15. Please review the discussion and include details of the implications for sexual health policy - what is missing and what needs to be developed and included based on the review? Similarly, what are the implications of the findings on practice and those seeking to develop and deliver such programmes in the future? Gaps in knowledge and confidence features in your review - what are the education and practice development needs that need to be addressed?
• The discussion has now been updated to include implications for policy. It reads as follows 'The findings of this review signal the requirement for sexual health policy and educational bodies to standardise an LGBTI+ inclusive curriculum for young people, ensuring facilitators are provided with adequate training and upskilling to deliver it effectively, and that this training and information is made accessible on a secure, national platform.'
• The discussion has now also been updated to include implications and needs for practice. It reads as follows 'These findings also have implications for practice, whereby the means by which sexual health education training is accessed by professionals will be fully online, removing any disparities between rural and central locations. This will also enable those accessing training to do so at times suitable to their individual schedules and feel confident in their ability to access information at any time for a reliable source as required.'

16. Please set out the gaps in research based on your analysis of the current evidence and the areas for future research and priorities.
• This has been updated as follows 'Further research is needed to better understand the ability of online training to address the need for inclusive sexual health education. Such research should examine the types of platforms sexual health information is delivered on, and how these are advertised, in order to reduce equality of access and information for young people and professionals. Research is also required in relation to the ability of online training to address the disparities that result due to negative facilitator attributes. Such research would help to further standardised inclusive sexual health education across school settings, reduce stigma and foster inclusion for LGBTI+ young people in the school environment.'

**VERSION 2 – REVIEW**

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| **REVIEWER** | Caroline Dorsen<br>Rutgers School of Nursing |

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| **REVIEW RETURNED** | 24-Feb-2021 |

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| **GENERAL COMMENTS** | Thank you for the opportunity to review this important and timely manuscript. Overall, I think that the review could use increased clarity around its aims and methods. As well, a few errors exist in the paper that *suggest* that a lack of knowledge around LGBTQIA+ culture and health (not meaning to be insulting, but the errors are noticeable and must be fixed in order for the paper to be seen as adding in a helpful, thoughtful and nuanced way to the literature).<br><br>1. Abstract. There are a number of errors and areas that need clarity in the abstract:<br>a. Should state: Qual, quant or mixed methods<br>b. Objective of systematic review is usually to critically appraise and synthesize, not just synthesize<br>c. Exclusion criteria: "studies based in philosophy or theory" is unclear<br>d. Conclusion: unclear if this is a systematic review about sex ed for LGBTQI youth, or about including LGBTQI content in sex education for all youth (if former, then inclusion criteria should be clearly state LGBTQ youth)<br><br>2. P. 4 are you sure that adolescent sexual activity has increased in past decades? Lots of reports of declining sexual activity among millennials.<br>3. P.4 line 16-17 conflating sexual orientation and gender identity. Discrete categories. Same lines 54-55. Not LGBTQI pronouns. They are gender pronouns<br>4. Clarity must be improved in many areas<br>5. Get rid of first paragraph?<br>6. Editing needed. Page 4, lines 30-31.<br>7. Search: not ok to just say range of data bases including. Need to be very specific so your readers can replicate your methods exactly.<br>8. Eligibility: nothing about LGBTQI in eligibility<br>9. Patient and public involvement: should explicitly state IRB exempt<br>10. Really nothing about limitations of studies themselves. But state did critically appraisal with CASP tools. What is the state of the literature? Table 1 gives a score 1-10 but doesn't help readers understand where limitations of the literature are.<br>11. Errors around LGBTQ health: page 11. Definition of bottom surgery is wrong. Not just for transfeminine people. |

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| **REVIEWER** | Todd Coleman<br>Wilfrid Laurier University |
| **REVIEW RETURNED** | 30-Mar-2021 |

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| **GENERAL COMMENTS** | Thank you for the opportunity to review this manuscript, and important work that focuses on the the characteristics of sexual health inclusive of LGBTI+ individuals. I noticed this is a revise and resubmit, which seems to have addressed most of the initial reviews. I would suggest some minor revisions that should be addressed prior to publication. These range in terms of the required work on the part of the co-authors. Comments are listed below, organized by manuscript section, labelled with the section they are found in.<br><br>Introduction (Page 3, Paragraph 1, line 7) "Sexuality is an activity of daily living [...]" This seems like an overly vague statement that still remains unclear. I would suggest deleting "an activity of daily living" and keep the remainder of the sentence. |

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| | Introduction (Page 3, Paragraph 1, line 15) "and all other sexualities/genders". The term LGBTI+ is not inclusive of all other sexualities or genders, as it does not typically include heterosexual/cissexual/cisgender people. I would suggest specifying that you're looking at minority sexual/gender identities.<br><br>Introduction (Page 3, Paragraph 3, line 35) A general stylistic comment about writing. Try to avoid the use of the word "surrounding" or "around" when you mean to write "about" (e.g. concerns can not 'surround' adolescent pregnancy).<br><br>Introduction (Page 3, Paragraph 3, line 42) There is a comma or bracket missing near "e.g.",<br><br>Introduction (Page 4, Paragraph 1, line 7) I believe you mean "perspective" rather than "prospective"?<br><br>Methods (Page 4, Study Eligibility, lines 42-54) While findings related to Ireland and Irish Law are important contextual factors to consider, these seem out of place in the context of a systematic review conducted on studies from around the world. I would suggest focusing a little more broadly on the role sexual health education plays in education policy, and how it might be viewed across several countries. This might be something you also bring into the Conclusion. Sexual health can be a very controversial issue in some areas, with many opinions put forward about what this should look like, depending on political parties in power, and the inclusion of LGBTI+ inclusive sexual health education has often been highlighted (i.e. parents outraged about having to learn something beyond a gender binary).<br><br>Methods (Page 4, Study Eligibility, line 58) Explain why articles/studies were not available in full text (e.g. was it due to library subscription, needing to pay for articles?).<br><br>Methods (Page 4, Study Eligibility, line 60) What in the abstract were you looking for to determine eligibility?<br><br>Methods (Page 12, Data Synthesis, line 29) Typos in the second line of this paragraph. I believe you were trying to say "a number of which were single cohort, pre-test post-test in design."<br><br>Results (overall) The headings and subheadings included here could be elucidated a bit more descriptively. For example, the main heading under 3.2 and the subheading "Online learning" seem very similar and should be distinguished from each other.<br><br>Results (Page 16, Online learning, line 21) "from" instead of "form".<br><br>Results (Page 16, Current Sexual Health Components, line 56) Missing word, "majority of sexual health education topics."<br><br>Results (Page 18, Paragraph 2, Line 15) How was "wellbeing" defined in these studies? This is a broad term and often means different things to different people.<br><br>Results (Page 18, Paragraph 2, Line 20) I believe there's some punctuation missing between words (e.g. care, equality, and intimacy, regardless of gender). |

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| | Results (Page 19, Paragraph 2, Line 24) Bottom surgery, while more commonly performed for trans feminine, is not exclusive to this group. A broader definition should be made here to be inclusive of the entire gender spectrum.<br><br>Results (Page 18, Negative Attributes, Line 27) Which LGBTI+ individuals are you referring to specifically, Intersex?<br><br>Discussion (Overall) Speaking back to the earlier comment on educational policy, suggestions are great, but this article should be considered in context of how difficult it is for inclusion of LGBTI+ sexual health education. Extreme pushback is common in many settings, and can derail and erase sexual and gender minorities, leaving young people with inadequate knowledge about themselves and their sexual health to make evidence-informed decision. A little more on the context would be helpful here.<br><br>Overall, this is a great article, and looking forward to seeing the final product. |

**VERSION 2 – AUTHOR RESPONSE**

**Reviewer: 1 [Dr. Caroline Dorsen, Rutgers School of Nursing]**

***Comments to the Author:***
***Thank you for the opportunity to review this important and timely manuscript. Overall, I think that the review could use increased clarity around its aims and methods. As well, a few errors exist in the paper that *suggest* that a lack of knowledge around LGBTQIA+ culture and health (not meaning to be insulting, but the errors are noticeable and must be fixed in order for the paper to be seen as adding in a helpful, thoughtful and nuanced way to the literature).***

- The paper has now been reviewed in its entirety.

1. ***Abstract. There are a number of errors and areas that need clarity in the abstract:***
   a. ***Should state: Qual, quant or mixed methods***
   b. ***Objective of systematic review is usually to critically appraise and synthesize, not just synthesize***
   c. ***Exclusion criteria: "studies based in philosophy or theory" is unclear***
   d. ***Conclusion: unclear if this is a systematic review about sex ed for LGBTQI youth, or about including LGBTQI content in sex education for all youth (if former, then inclusion criteria should be clearly state LGBTQ youth)***

   - The abstract has now been edited to reflect these comments.

2. **P. 4 are you sure that adolescent sexual activity has increased in past decades? Lots of reports of declining sexual activity among millennials.**
   - Many thanks to the reviewer for this interesting viewpoint. In line with both reviewer comments the authors have now removed the first paragraph and edited the introduction to ensure it is reflective of the manuscript.

3. ***P.4 line 16-17 conflating sexual orientation and gender identity. Discrete categories. Same lines 54-55. Not LGBTQI pronouns. They are gender pronouns***
   - This has now been updated.

4. ***Clarity must be improved in many areas***
   - The manuscript has been edited throughout

5. ***Get rid of first paragraph?***
   - This paragraph has now been removed.

6. ***Editing needed. Page 4, lines 30-31.***
   - This has now been edited.

7. ***Search: not ok to just say range of data bases including. Need to be very specific so your readers can replicate your methods exactly.***
   - This has now been edited to read *"A comprehensive literature search was conducted utilising the databases including EMBASE, PsychINFO and SocINDEX."*

8. ***Eligibility: nothing about LGBTQI in eligibility***
   - Many thanks to the reviewer for this comment. Eligibility is now updated to reflect same and reads *"if they were (i) peer reviewed journal articles; (ii) related to inclusive sexual health in an educational or online setting; (iii) focused on training or educating. English language content was chosen to ease assessment by a native English-speaking researcher"*

9. ***Patient and public involvement: should explicitly state IRB exempt***
   - This section has been edited to *include "Reviews are exempt from Institutional Review Boards."*

10. ***Really nothing about limitations of studies themselves. But state did critically appraisal with CASP tools. What is the state of the literature? Table 1 gives a score 1-10 but doesn't help readers understand where limitations of the literature are.***
    - Thanks to the reviewer for this comment. In order to keep the information in the table accessible we have updated the limitations section to read: *"Including studies of varying quality could yield misleading results. In terms of the included studies a number of limitations were noted. The majority did not discuss the relationship between the researcher and the participants. In addition, pre-defined sample size calculations were not reported in all studies."*

11. ***Errors around LGBTQ health: page 11. Definition of bottom surgery is wrong. Not just for transfeminine people.***
    - Thanks to the reviewer for this comment. The section now reads: *"binding and 'bottom surgery' (this is a form of gender affirmation surgery)"*

**Reviewer: 2 [Dr. Todd Coleman, Wilfrid Laurier University]**

***Comments to the Author:***
***Thank you for the opportunity to review this manuscript, and important work that focuses on the the characteristics of sexual health inclusive of LGBTI+ individuals. I noticed this is a revise and resubmit, which seems to have addressed most of the initial reviews. I would suggest some minor revisions that should be addressed prior to publication. These range in terms of the required work on the part of the co-authors. Comments are listed below, organized by manuscript section, labelled with the section they are found in.***

1. ***Introduction (Page 3, Paragraph 1, line 7) "Sexuality is an activity of daily living [...]" This seems like an overly vague statement that still remains unclear. I would suggest deleting "an activity of daily living" and keep the remainder of the sentence.***
   - In line with the first reviewers comments, this paragraph has now been removed.

2. ***Introduction (Page 3, Paragraph 1, line 15) "and all other sexualities/genders". The term LGBTI+ is not inclusive of all other sexualities or genders, as it does not typically include heterosexual/cissexual/cisgender people. I would suggest specifying that you're looking at minority sexual/gender identities.***
   - In line with the first reviewers comments, this paragraph has now been removed.

3. ***Introduction (Page 3, Paragraph 3, line 35) A general stylistic comment about writing. Try to avoid the use of the word "surrounding" or "around" when you mean to write "about" (e.g. concerns can not 'surround' adolescent pregnancy).***
   - Thanks for this constructive comment. Manuscript edited to reflect same.

4. ***Introduction (Page 3, Paragraph 3, line 42) There is a comma or bracket missing near "e.g.",***

- Thanks has now been edited.

5. ***Introduction (Page 4, Paragraph 1, line 7) I believe you mean "perspective" rather than "prospective"?***
   - Thanks has now been edited.

6. ***Methods (Page 4, Study Eligibility, lines 42-54) While findings related to Ireland and Irish Law are important contextual factors to consider, these seem out of place in the context of a systematic review conducted on studies from around the world. I would suggest focusing a little more broadly on the role sexual health education plays in education policy, and how it might be viewed across several countries. This might be something you also bring into the Conclusion. Sexual health can be a very controversial issue in some areas, with many opinions put forward about what this should look like, depending on political parties in power, and the inclusion of LGBTI+ inclusive sexual health education has often been highlighted (i.e. parents outraged about having to learn something beyond a gender binary).***
   - Thanks for this constructive comment. The authors discussed same in detail and have reformatted the manuscript to incorporate this into the introduction and conclusion, highlighting how the results may support policy makers internationally.

7. ***Methods (Page 4, Study Eligibility, line 58) Explain why articles/studies were not available in full text (e.g. was it due to library subscription, needing to pay for articles?).***
   - The text has now been edited and reads *"Articles were excluded from the review if: (i) the population was not LGBTI+ inclusive; (ii) studies were based in philosophy or theory and/or (iii) the study was not available in full text (due to institution library subscriptions)."*

8. ***Methods (Page 4, Study Eligibility, line 60) What in the abstract were you looking for to determine eligibility?***
   - Many thanks to the reviewer for this comment. As with Reviewer 1, eligibility has now been updated and reads *"if they were (i) peer reviewed journal articles; (ii) related to inclusive sexual health in an educational or online setting; (iii) focused on training or educating. English language content was chosen to ease assessment by a native English-speaking researcher"*

9. ***Methods (Page 12, Data Synthesis, line 29) Typos in the second line of this paragraph. I believe you were trying to say "a number of which were single cohort, pre-test post-test in design."***
   - Thanks, this has now been updated.

10. ***Results (overall) The headings and subheadings included here could be elucidated a bit more descriptively. For example, the main heading under 3.2 and the subheading "Online learning" seem very similar and should be distinguished from each other.***
    - This comment was very helpful. We have now reviewed the headings and subheadings as a group, in line with the manuscript and analysis and edited the manuscript.

11. ***Results (Page 16, Online learning, line 21) "from" instead of "form".***
    - Thanks, this has now been updated.

12. ***Results (Page 16, Current Sexual Health Components, line 56) Missing word, "majority of sexual health education topics."***
    - Thanks, this has now been updated.

13. ***Results (Page 18, Paragraph 2, Line 15) How was "wellbeing" defined in these studies? This is a broad term and often means different things to different people.***
    - Thanks to the reviewer for highlighting this issue. Upon review the section has been updated in line with the referenced author view of sexual and emotional wellbeing and reads as follows: "*The ability to access sexual health information online was shown to result in increased sexual and emotional wellbeing for LGBTI+ youth, who report poorer levels of sexual health education and higher levels of dissatisfaction with their sex lives"*

14. ***Results (Page 18, Paragraph 2, Line 20) I believe there's some punctuation missing between words (e.g. care, equality, and intimacy, regardless of gender).***

- Thanks, this has now been updated.

15. ***Results (Page 19, Paragraph 2, Line 24) Bottom surgery, while more commonly performed for trans feminine, is not exclusive to this group. A broader definition should be made here to be inclusive of the entire gender spectrum.***
    - Thanks, this has now been updated.

16. ***Results (Page 18, Negative Attributes, Line 27) Which LGBTI+ individuals are you referring to specifically, Intersex?***
    - The text has now been updated for clarity and reads *"Facilitators were also shown to mainly deliver sexual health from a binary perspective (male and female) excluding some LGBTI+ youth including non-binary and intersex communities"*

17. ***Discussion (Overall) Speaking back to the earlier comment on educational policy, suggestions are great, but this article should be considered in context of how difficult it is for inclusion of LGBTI+ sexual health education. Extreme pushback is common in many settings, and can derail and erase sexual and gender minorities, leaving young people with inadequate knowledge about themselves and their sexual health to make evidence-informed decision. A little more on the context would be helpful here.***
    - Thanks, in line with your earlier comment this has now been updated.

***Overall, this is a great article, and looking forward to seeing the final product.***
- Thank you.